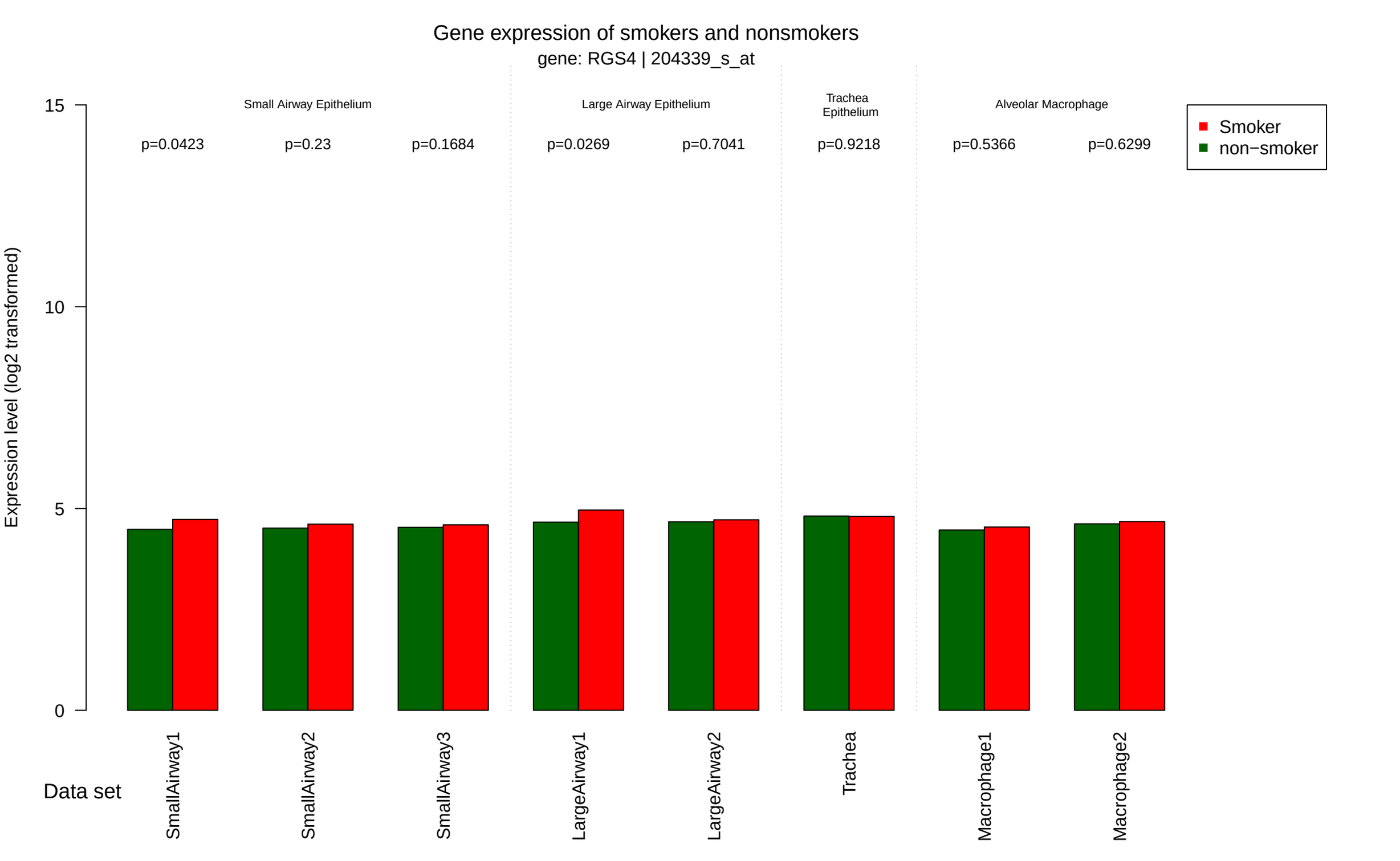
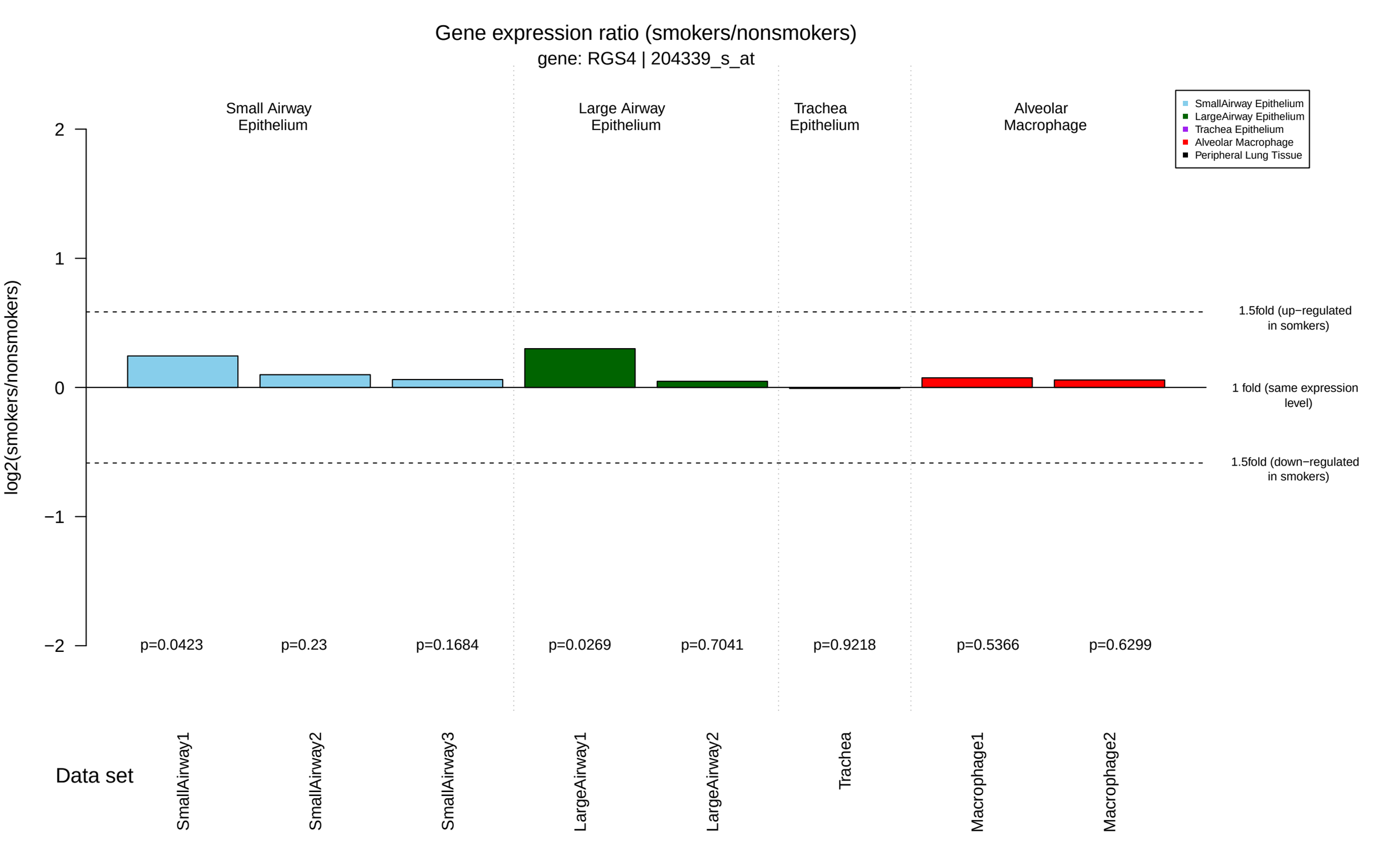
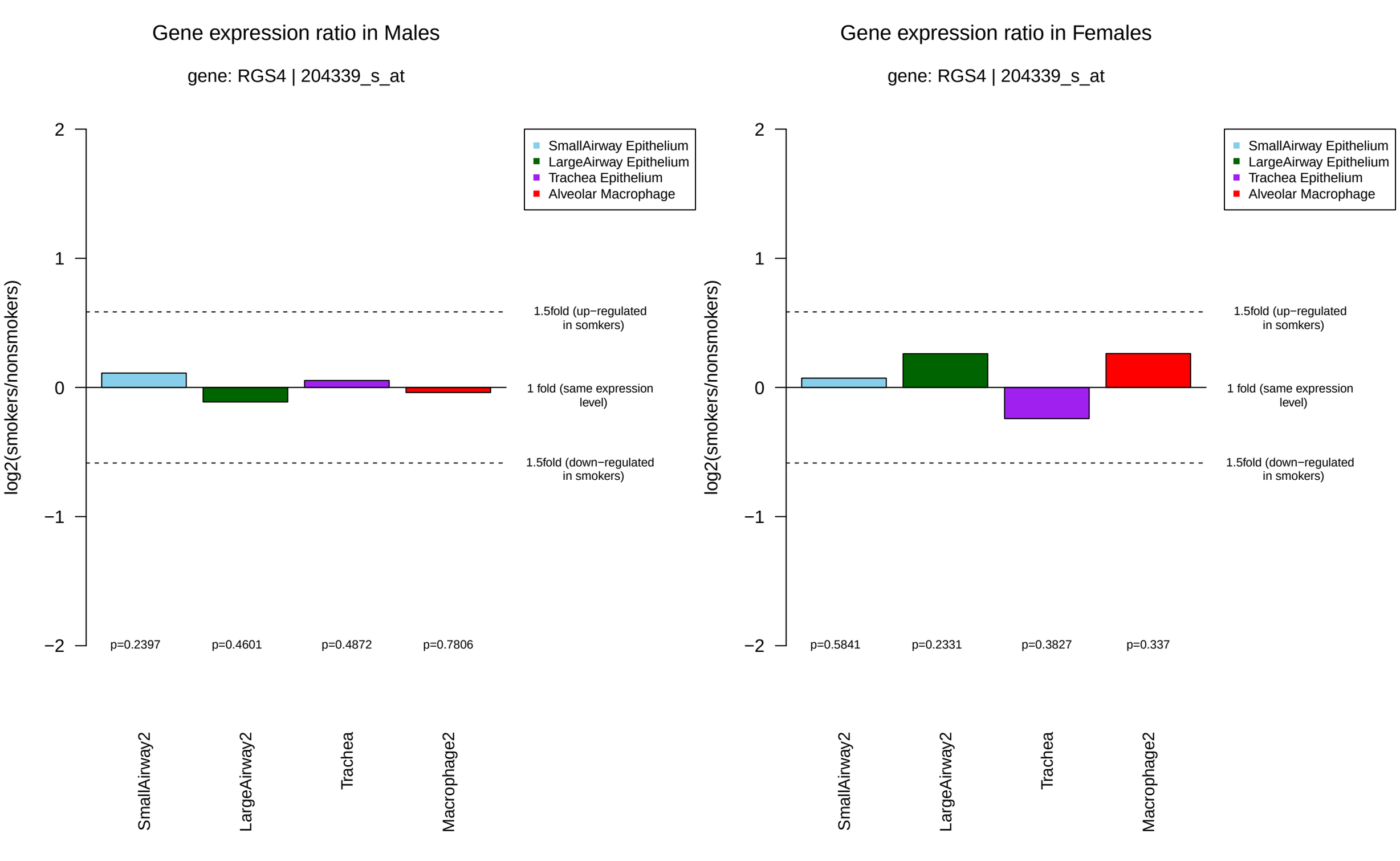
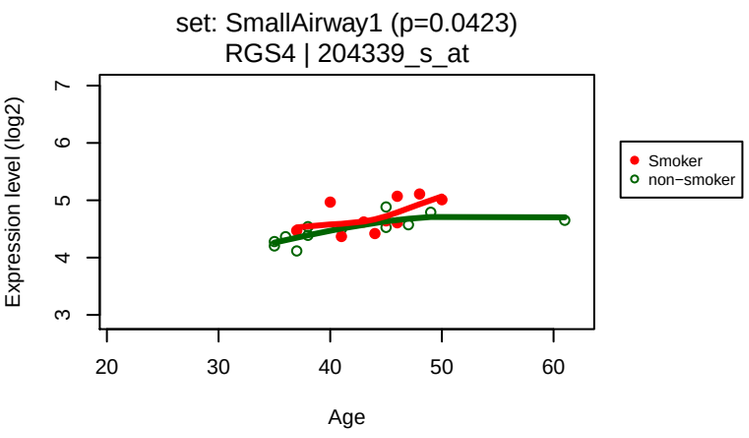
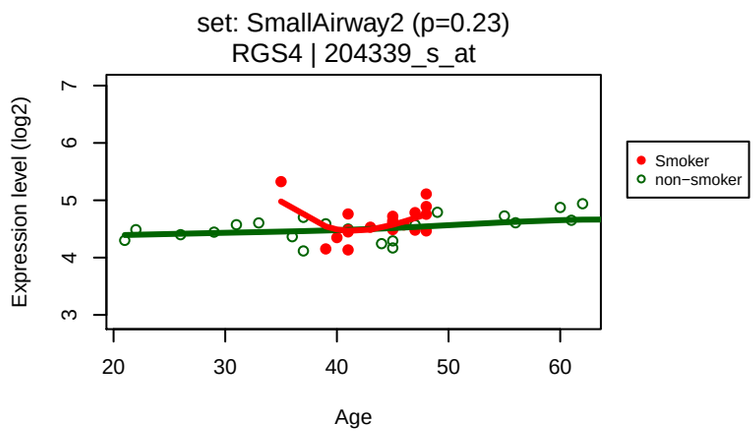
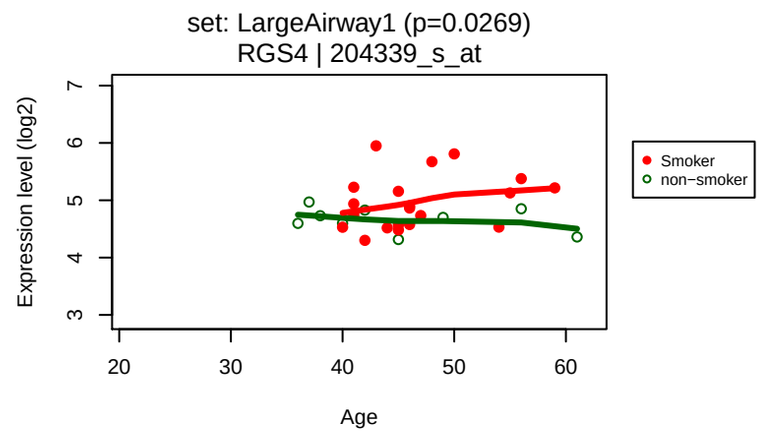

Gene expression of smokers and nonsmokers
gene: RGS4 | 204339_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0423
p=0.23
p=0.1684
p=0.0269
p=0.7041
p=0.9218
p=0.5366
p=0.6299
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

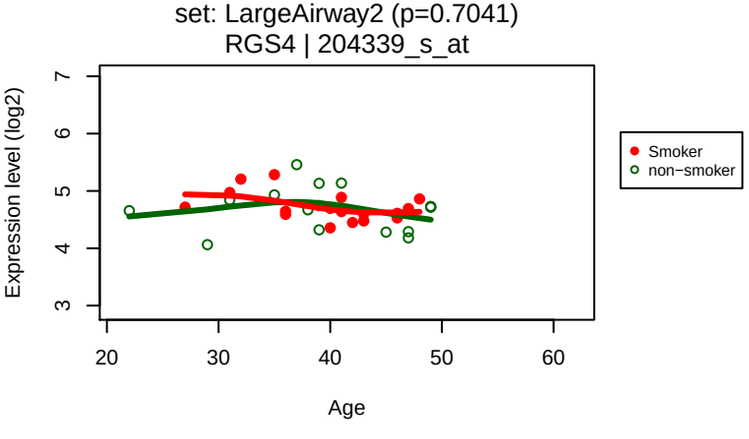

Gene expression ratio (smokers/nonsmokers)
gene: RGS4 | 204339_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0423
p=0.23
p=0.1684
p=0.0269
p=0.7041
p=0.9218
p=0.5366
p=0.6299
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

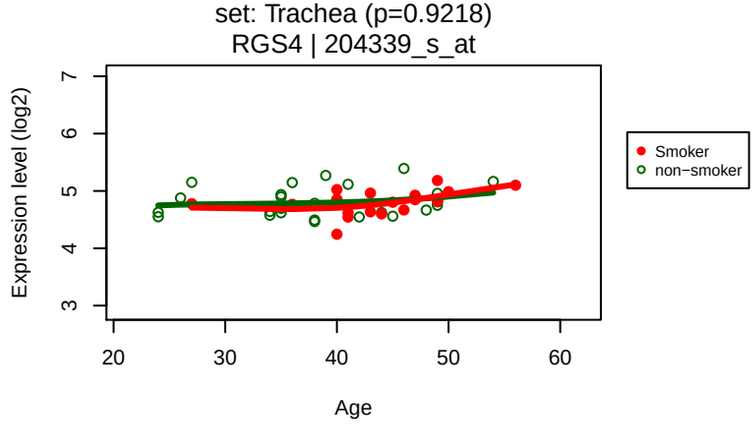

Gene expression ratio in Males
gene: RGS4 | 204339_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2397
p=0.4601
p=0.4872
p=0.7806
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS4 | 204339_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5841
p=0.2331
p=0.3827
p=0.337
SmallAirway2
LargeAirway2
Trachea
Macrophage2

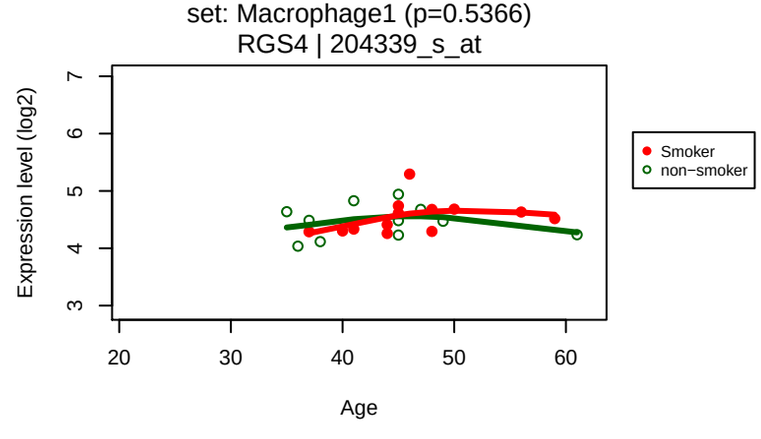

set: SmallAirway1 (p=0.0423)
RGS4 | 204339_s_at
Expression level (log2)
Age
Smoker
non−smoker

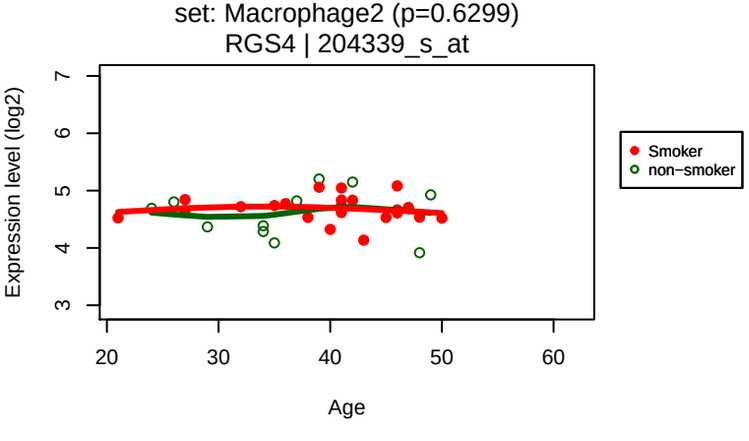

set: SmallAirway2 (p=0.23)
RGS4 | 204339_s_at
Expression level (log2)
Age
Smoker
non−smoker

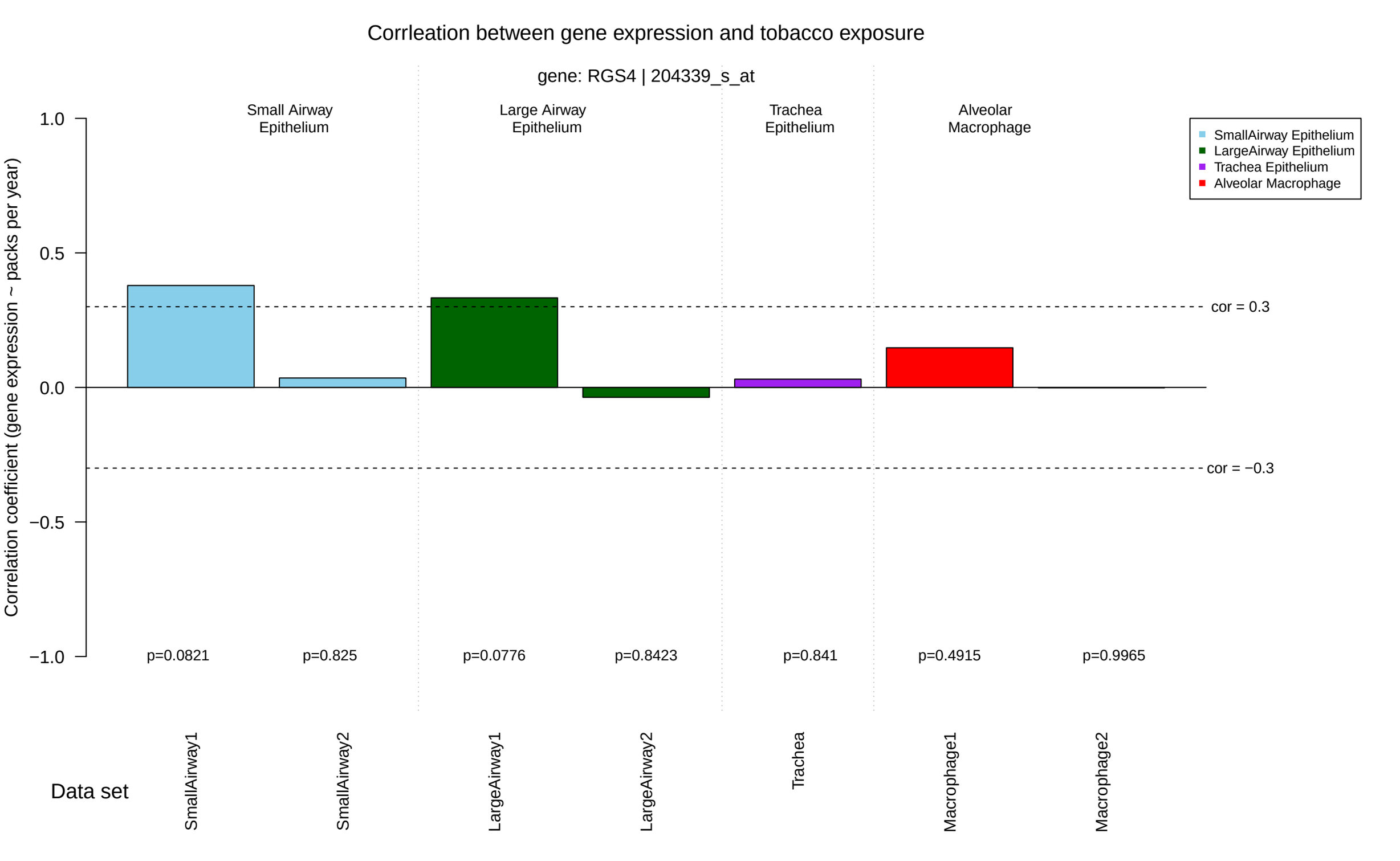

set: LargeAirway1 (p=0.0269)
RGS4 | 204339_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7041)
RGS4 | 204339_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9218)
RGS4 | 204339_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5366)
RGS4 | 204339_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6299)
RGS4 | 204339_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS4 | 204339_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0821
p=0.825
p=0.0776
p=0.8423
p=0.841
p=0.4915
p=0.9965
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2